

Testing BISG Race Imputation Against Known Race

BISG checked against the truth, on forty-six thousand Georgia voters

84-355 Democracy's Data

August 2026

A statistical inference cannot normally be checked. That is what makes it an inference rather than a measurement: if the answer were already written down, nobody would need to estimate it.

Georgia writes it down. Race appears on the voter registration form, collected for Voting Rights Act compliance rather than for anybody's research. So in one Georgia county there are 45,984 people whose race a widely used statistical method is trying to guess, and whose race is already on file.

The method is **Bayesian Improved Surname Geocoding**, or BISG. **Elliott and colleagues (2009)** built it to estimate a patient's race from a surname and a neighbourhood, for health records that did not collect it. **Imai and Khanna (2016)** brought it to voter files, which is how it reached redistricting cases. Here it can be graded.

So: **how good is the guess?**

It sounds like a question with one answer. It has five, and they disagree with each other by more than seventy points.

01 Why anybody needs to know

BISG estimates who the voters are. Ecological inference then estimates how those groups voted. Chained together, the two are the standard evidentiary basis of a Section 2 vote-dilution case — the evidence on which districts are drawn or struck down. The same method is used by the Consumer Financial Protection Bureau to detect lending discrimination.

In both settings the error rate is almost never reported, because in both settings it is almost never knowable. A federal court is handed an estimate with no accompanying statement of how often the method is wrong, for whom, or in which direction.

Nobody serious argues that BISG should therefore be thrown out, because the alternative is no evidence at all. The dispute is about **what has to accompany an estimate before a court should act on it**. And about whether one overall accuracy figure is a fair summary of a method whose performance varies enormously by group. This county is one of the few places where the aggregate figure can be taken apart and the pieces looked at.

02 An answer key that exists for an unrelated reason

Three files meet in this chapter and none of them was built to meet the others. The first is a Georgia voter registration extract for Houston County, carrying the race each voter wrote on the form. It comes from the working data assembled for the county's 2026 redistricting matter. The second is the 2020 census count of who lives on each of the county's blocks. The third is the Census Bureau's list of how often each surname is held by people of each race, tabulated from the 2010 census and published for exactly this kind of use. It is read here out of the surnames chapter's folder rather than copied, so the two chapters cannot disagree about it.

The registration file is the one that makes the chapter possible, and it exists for a reason that has nothing to do with research. A small number of states put a race question on the voter registration form, most of them states once subject to federal preclearance. The Voting Rights Act obliged them to be able to report who was registering. Nobody collected it to grade a statistical method. But the effect is that in this county the thing BISG is trying to estimate is already written down, which almost nowhere else it is.

That is also the deepest problem with the whole literature, and it is a problem about measurement rather than about this file. **An inference can only be graded where the answer is already recorded, and the answer is only recorded where the inference is unnecessary.** Every published accuracy figure for BISG therefore comes from the handful of places that did not need BISG — and the method is used overwhelmingly in the places that do. Carrying a number from here to there is an assumption about how alike two electorates are, and no amount of care with this county's data converts it into a measurement. What would have been better is a national file with the same truth column, which does not exist and, for good reasons, is not going to.

One decision was made before anything else and it is the one that matters most. The source registration file carries a flag marking records whose race was not supplied by the voter but **filled in by BISG itself**, and every one of those records is dropped here. Scoring a method against its own output would have produced a beautiful and completely meaningless result, and it would have looked exactly like a success.

The rest of the cleaning is smaller. The extract arrives with dozens of columns, including full name, birth year and street address. Three are carried forward: surname, reported race, census block. Nothing identifying is committed here at all.

And the census side got a check worth copying. The script requires the white-alone-not-Hispanic column of one census table to match the same column of the other, person for person, and stops if it does not. It matches. The Black column does not, and that mismatch is how the any-part definition used throughout this chapter announced itself rather than being assumed.

The three files also arrived three different ways. The surname list was published by the Census Bureau for exactly this use, and it is an ordinary download. It is not downloaded twice: this chapter reads the copy already sitting in the surnames chapter's folder. The block counts are the Bureau's own tables, public on the same terms.

The registration extract came from neither. It came out of a case file: the working data assembled for the 2026 Houston County matter, on one machine, behind no URL at all. It is read from that disk, and the work stops if it is not there. So for anyone but its author, the CSV here is not the output of a build that can be re-run. It is the only copy there is.

Something arrived with it that is easy to miss. The block each voter sits in came already assigned. Somebody matched street addresses to census blocks before this chapter

ever saw the file. Every address that failed to match, or matched to the wrong side of a street, was settled by a judgment no column records. Geography is half of what the method below uses. Grading BISG here means grading it partly on the quality of a geocode nobody here performed.

03 Fifty-three columns arrive, and three survive

The registration extract cannot be shown here, and that is the point of the section rather than an obstacle to it. What can be shown is its shape. These are the column names, exactly as they arrive on the first line of the file:

Position	Column as it arrives	What it holds	This chapter
1	County	the county that maintains this registration	—
2	Voter Registration Number	the state's identifier for this registration	—
3	Status	active or inactive — a legal status, not a description of a person	—
4	Status Reason	why the status is what it is	—
5	Last Name	surname — one of the two columns this chapter is allowed to want	kept
6	First Name	given name	—
7	Middle Name	middle name	—
8	Suffix	Jr, III, and so on	—
9	Birth Year	year of birth only; Georgia does not publish the day	—
10	Residence Street Number	house number	—
11	Residence Pre Direction	the N in N Main St	—
12	Residence Street Name	street name	—
13	Residence Street Type	St, Ave, Rd	—
14	Residence Post Direction	the NW in Main St NW	—
15	Residence Apt Unit Number	apartment or unit	—
16	Residence City	city	—
17	Residence Zipcode	ZIP	—
18	County Precinct	the precinct code this voter votes in	—
19	County Precinct Description	that precinct's name in words	—
20	Municipal Precinct	city precinct code, where the city runs its own	—
21	Municipal Precinct Description	that precinct's name in words	—
22	Congressional District	U.S. House district	—
23	State Senate District	Georgia Senate district	—

	Position	Column as it arrives	What it holds	This chapter
24		State House District	Georgia House district	—
25		Judicial District	judicial district	—
26		County Commission District	county commission district	—
27		School Board District	school board district	—
28		City Council District	city council district	—
29		Municipal School Board District	city school board district	—
30		Water Board District	water board district	—
31		Super Council District	a Georgia 'super' district, overlaying several ordinary ones	—
32		Super Commissioner District	the same idea, for the county commission	—
33		Super School Board District	the same idea, for the school board	—
34		Fire District	fire district	—
35		Municipality	city, where the address is in one	—
36		Combo	a Georgia term for the unique combination of districts an address falls in — the ballot style	—
37		Land Lot	colonial-era land survey lot, still carried in Georgia records	—
38		Land District	the survey district that lot sits in	—
39		Registration Date	when this registration was made	—
40		Race	race as recorded by the state — the other column this chapter wants	kept
41		Gender	sex as recorded by the state	—
42		Last Modified Date	when the record last changed	—
43		Date of Last Contact	when the county last had contact with this voter	—
44		Last Party Voted	the last primary ballot taken — Georgia has no party registration	—
45		Last Vote Date	when this voter last voted	—
46		Voter Created Date	when the record was created, which is not the registration date	—
47		Mailing Street Number	mailing address, where it differs from residence	—
48		Mailing Street Name	mailing address	—
49		Mailing Apt Unit Number	mailing address	—
50		Mailing City	mailing address	—
51		Mailing Zipcode	mailing address	—
52		Mailing State	mailing address	—
53		Mailing Country	mailing address	—

53 columns and 127,560 rows, one per registered voter in the county. Not one of those rows appears anywhere in this book. The two columns marked with a star are the only ones this chapter is allowed to want. Everything else is either irrelevant to a surname model, or the kind of thing that must not be carried into a folder handed to a class. A first

name, a middle name, a birth year, a street number, a street name, an apartment number, a mailing address in another state.

The race column is not in that file at the point the chapter uses it. It comes from a second file, one row per voter, whose relevant columns are `race_self`, `race_raw`, `GEOID20` and a flag called `is_bisg_imputed`. That flag is the whole chapter in one column.

The cleaned result has three columns and no way back to a person:

Surname	Race the voter reported	Census block
AALTONEN	white	131530211253020

Four judgments got from 53 columns to three, and every one of them changes what the chapter can claim.

Drop the rows where the answer key was written by the thing being graded. 11,168 of the 155,756 records in the race file have `is_bisg_imputed` set true: the voter left the question blank and somebody filled it in by running BISG. Score BISG against those rows and it looks superb. Nothing in the numbers would give the circularity away, because a model checked against its own output agrees with itself. They are dropped first, before anything else happens.

Decide what counts as an answer. A further 854 voters reported a race the five-category scheme has no box for, or none at all. They are dropped too, which is not a neutral act: the method is now being graded only on people who fit the categories it predicts.

Throw away the identifiers on purpose. The registration number is what joins the two files, and it is discarded the moment the join is done. The committed file has no key, which means it can never be linked back to a person and also that nobody can re-derive it or audit it against the source. That is a deliberate trade, and it is the same trade the chapter on levels of aggregation is about.

Accept a geocode you did not perform. `GEOID20` arrives already filled in. Somebody matched a street address to a census block before this chapter existed, and every address that failed, or landed on the wrong side of a street, was resolved by a rule no column records. Half of what BISG uses is geography, so part of what gets graded below is the quality of that geocode.

What comes out is 49,771 rows across 2,130 blocks, and the census tables that go with it are aggregated to the same 3,269 blocks so that the two can be joined on a fifteen-character string.

04 What one record is

Surname	Race the voter reported	Census block
AARON	asian	131530212041012
AARON	black	131530203021001
AARON	white	131530202003009

Three columns: a surname, the race the voter reported on the registration form, and the census block they live in. **No names and no addresses** — the model only ever receives the surname and the block, so that is all the file carries.

Quantity	Value
Registered voters in the file	49,771
Distinct census blocks they live in	2,130
Blocks in the county	3,269
County population by race (from the 2020 census)	white 86,211, Black 56,520, Hispanic 11,807, Asian 5,029, Am. Indian 339

The block file is the **geographic prior**: for each block, who lives there.

05 What gets dropped before anything is measured

Quantity	Value
Voters in the file	49,771
Whose surname appears in the Census surname list	45,984 (92.4%)
Whose surname does not	3,787
Voters the method is scored on	45,984

3,787 voters, 7.6% of them, hold a surname that is not in the Census list, because fewer than a hundred Americans shared it in 2010. **The method has no opinion about them at all**, so they are not in any figure below. Who they are turns out to matter a great deal.

06 The surname alone

The voter actually is	How many	Surname guesses right
white	23,668	94.5%
Black	16,841	16.9%
Hispanic	2,319	71.4%
Asian	2,038	64.2%
Am. Indian	1,118	0.2%

Overall accuracy: 61.3%.

That is the number a methods section reports, and it sounds like a method that works.

Before you read on, write down three numbers. The surname alone scores 61.3% overall. First: what percentage of *white* voters do you think it gets right? Second: what percentage of *Black* voters? Third, and this is the one that matters – **what would a method score if it simply guessed “white” for every single voter in this county**, with no surname, no geography, and no model at all?

07 Read down the column, not across the bottom

The accuracy column in that table is the thing to look at, and it is not what the overall figure suggests.

The surname gets white voters right 94.5% of the time and Black voters right 16.9% of the time. And:

Method	Overall accuracy
Surname only	61.3%
Guess ‘white’ for everybody	51.5%
Difference	+9.8 points

Guessing “white” for every voter scores 51.5%, because 51.5% of these voters are white. Almost nobody writes that number down; the guess is usually somewhere near a fifth, on the reasonable-sounding intuition that a five-category problem gives a coin-flip method about one chance in five. The floor is not one in five. It is a little over half, and it is available to anyone willing to type one word.

Here is the county, one square per percent of the 45,984 voters being scored.

guessing "white" every time collects 51.5% of the electorate

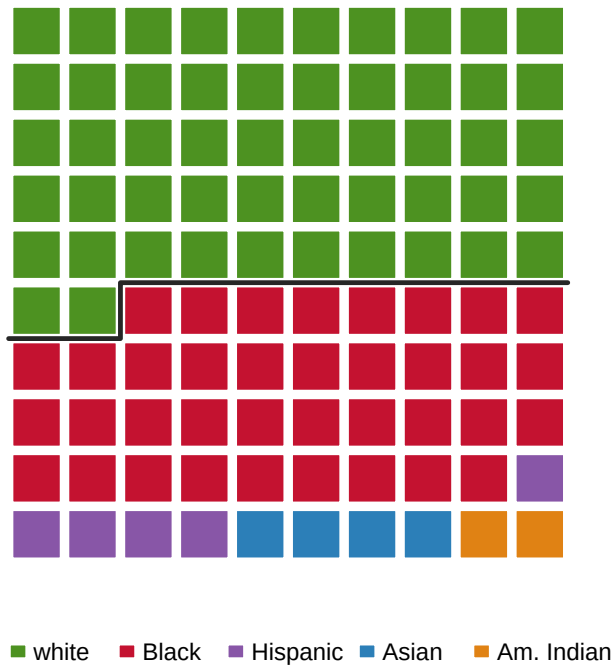


Figure 1. The electorate being scored, one square per percent. Each of the 100 squares is one percent of the 45,984 voters the method is graded on, colored by the race the voter reported and allocated by largest remainder. The heavy line encloses the green squares: 51.5 of them, which is what a method with no method at all collects by guessing "white" every time. The surname's 61.3% has to be read against that line, not against zero.

The overall figure is not a summary of how the method performs. It is a summary of who lives in Houston County. **On an unbalanced problem, overall accuracy mostly reports the balance.**

08 Geography is a trade, not an improvement

Surname alone is not BISG. The method multiplies the surname distribution by the block composition, which is the step Elliott and colleagues (2009) added. If a name is ambiguous but the block is overwhelmingly Black, the block should carry the inference.

The voter actually is	How many	Surname only	Surname + block	Change
white	23,668	94.5%	90.6%	-3.9
Black	16,841	16.9%	40.7%	+23.8
Hispanic	2,319	71.4%	65.9%	-5.5
Asian	2,038	64.2%	55.3%	-8.8
Am. Indian	1,118	0.2%	0.3%	+0.1

Method	Overall accuracy
Surname only	61.3%
BISG: surname + block	67.3%

Overall accuracy rises from 61.3% to 67.3% — +6.0 points. That is the number a methods section would report, and it is the wrong number.

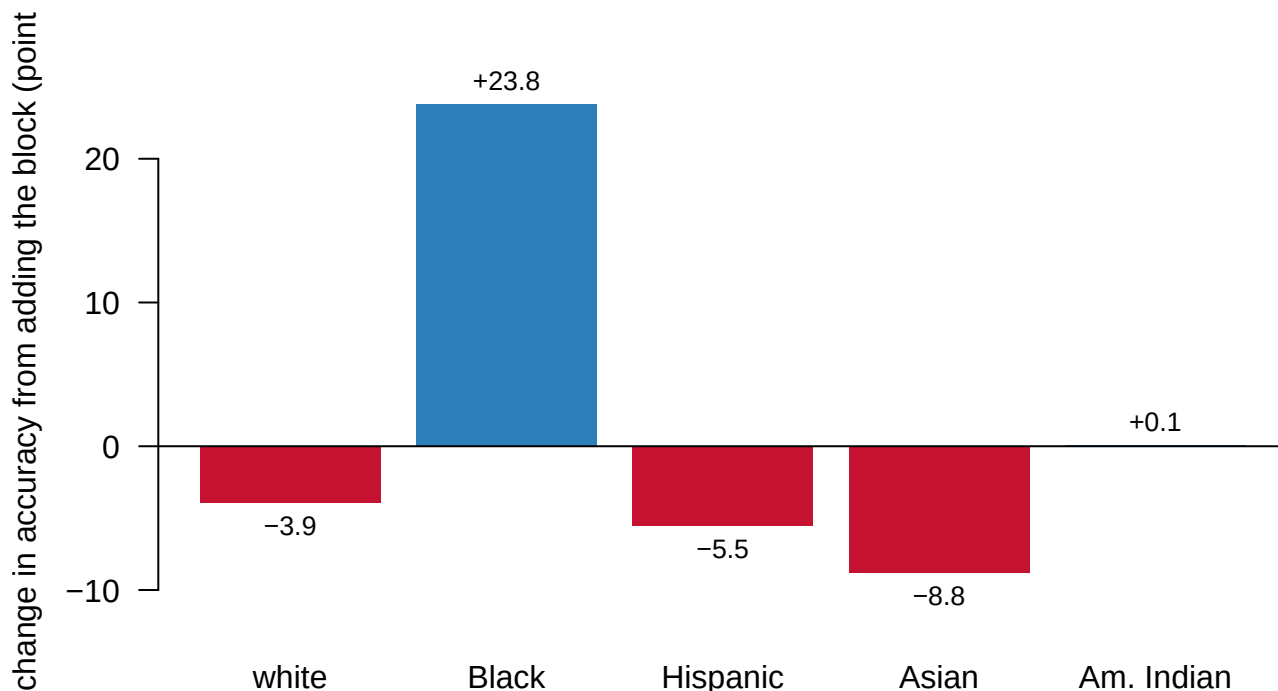


Figure 2. What adding the census block does to each group, separately. The change in accuracy from multiplying the surname distribution by the block's racial composition, in percentage points. One bar rises and three fall. The +6.0-point gain in the overall figure is the sum of these movements weighted by group size, which is why it reports none of them.

Adding geography is not a free improvement. It is a redistribution. Black voters gain +23.8 points, from 16.9% to 40.7%. That is the method working as designed, and it works because Black residents of this county are geographically concentrated, so the block genuinely carries information about them. Asian voters lose 8.8 points, Hispanic voters 5.5, white voters 3.9. Those populations are spread thinly across the county, so no block is distinctively Asian or Hispanic; the geographic term drags the estimate toward the local majority and overrides a surname signal that was already decent.

One group gained 23.8 points. Three lost. The +6.0 overall conceals all of it, and a report that quotes only the overall figure has not lied about anything.

And the number that should stay with you: with everything the method has, BISG identifies Black voters correctly **40.7% of the time**. It is wrong about them more often than it is right — in a county that is about a third Black, on the question Section 2 litigation turns on.

An accuracy column reports only the diagonal. Being wrong is not one thing; the errors go somewhere, and where they go is the rest of the table.

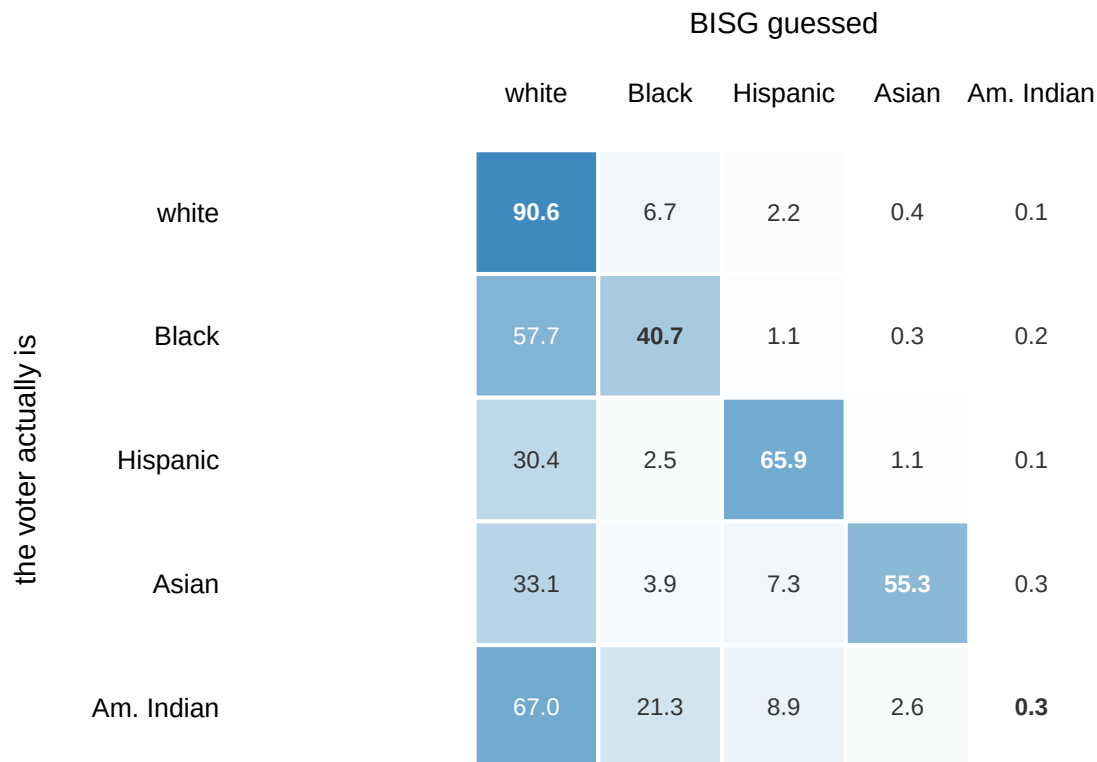


Figure 3. Where the errors go. Each row is a group of voters as they reported themselves; each column is what BISG guessed; rows sum to 100%. The outlined diagonal is the accuracy column from the table above, and everything off it is a specific mistake. The largest single off-diagonal cell is 67.0% of Am. Indian voters labeled white; the second column of the Black row shows 57.7% of Black voters labeled white. **The errors are not scattered.** They run in one direction, toward the group that is locally largest, which is exactly what multiplying by a geographic prior does.

09 Two places the method has no answer

Situation	Voters
Voters living in a block the census records as having nobody in it	392
Voters whose surname and block flatly contradict each other	2
Total needing a fallback	394
of whom are Black	158

392 voters live in a block the census records as empty, mostly new construction. The geographic term vanishes for them, and the county's own composition is used instead. And for 2 voters the surname is 100% one group while their block has zero residents of that group, so every category multiplies to zero. **BISG is undefined for them.**

Without those two lines of handling, R would raise no error. The routine that picks the largest of the final probabilities returns the *first* column on an all-zero row, so all 394 would have been silently labeled **white** — and 158 of them are Black.

A quiet default is worse than a crash.

10 Confidence, and the decision rule

Accuracy asks whether the winning label was right. Ask instead how much probability the model put on the *true* answer.

The voter actually is	Mean probability BISG placed on the truth	How often BISG's top guess was right
white	0.79	90.6%
Black	0.43	40.7%
Hispanic	0.60	65.9%
Asian	0.54	55.3%
Am. Indian	0.01	0.3%

A mean hides a shape. Here is the whole distribution — for every voter, how much probability BISG put on the answer that turned out to be correct.

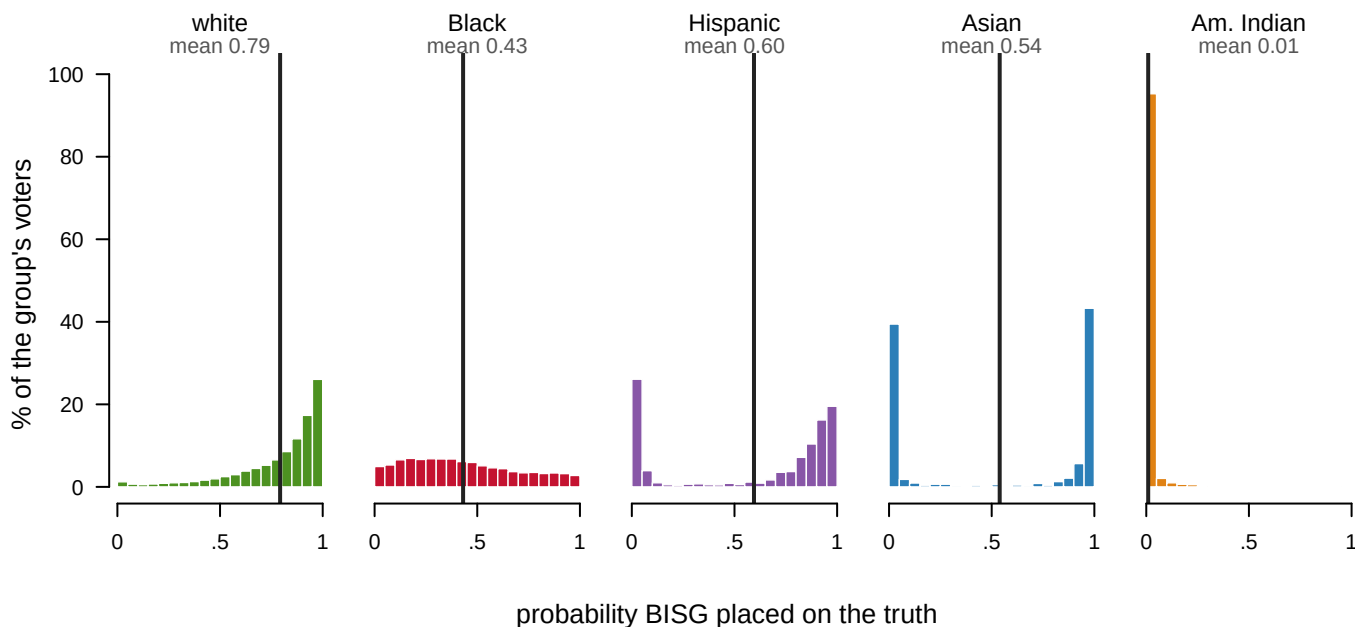


Figure 4. How much probability the model placed on the answer that was correct. One panel per group, on a common vertical scale, each bar a five-point slice of probability; the vertical rule in each panel is that group's mean. White voters pile against the right-hand wall at a mean of 0.79. Am. Indian voters pile against the left-hand wall at 0.01. The Black panel is the one to sit with: it is not concentrated at either end but spread through the middle, at a mean of 0.43.

For Black voters the model puts 0.43 on the truth. **It is not confidently wrong; it is genuinely uncertain** — and then the decision rule forces a hard guess anyway. A meaningful part of the information loss is in collapsing a probability to a label, not in the model.

For American Indian voters the model puts 0.01 on the truth. Those 1,118 voters cannot be found by any threshold at all. Recalibration would not fix that.

11 Two checks that cut the other way

First: does geography help, or does *this* geography help? Replace the block prior with a single county-wide one, so geography carries no information about where within the county anybody lives.

Prior	Overall	Black voters
Block-level (BISG)	67.3%	40.7%
Surname only, no geography	61.3%	16.9%
County-level prior	58.8%	7.9%

A county prior scores **worse than no prior at all**. It multiplies every voter by the same composition, which is 51.5% white, so it tilts everyone toward white and destroys information the surname had. “We controlled for geography” means nothing without saying at what scale.

Second: how does it do on the people it is actually for? Restrict to voters whose surname gives no group above 60% — the ambiguous names.

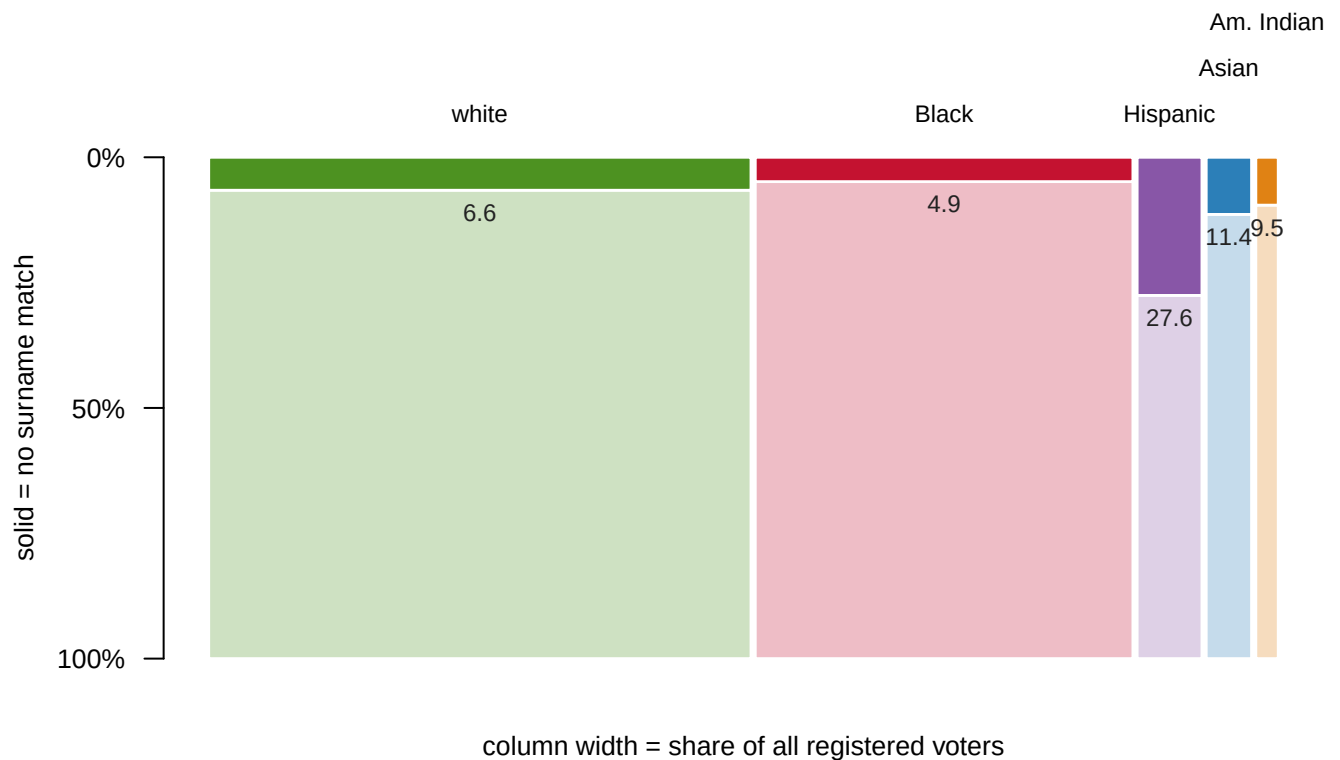
Quantity	Value
Voters with an ambiguous surname	8,450
Surname only, on those voters	43.3%
BISG, on those voters	59.4%

+16.1 points — the largest gain in any *overall* accuracy figure anywhere in this chapter, more than twice the +6.0 it gains on the full file. (The single largest gain of any kind is still the +23.8 points for Black voters above.) **The method is not uniformly weak**. It is weak where the surname was already informative and strong where it was not, which is the opposite of the impression Figure 2 leaves on its own. Both are true, and a fair account has to carry both.

12 Who never entered the test at all

Group	Share of all registered voters	Share of the voters with no surname match
white	50.9%	44.0%
Black	35.6%	22.7%
Hispanic	6.4%	23.3%
Asian	4.6%	6.9%
Am. Indian	2.5%	3.1%

Two numbers are fighting in that table — how big each group is, and how often it is dropped. Give each group a column as wide as it is large, and shade the part of the column the surname list has no opinion about.



Every rectangle is proportional to the number of people in it. The white column is the widest (50.9% of all voters) and only 6.6% solid; the Hispanic column is 6.4% of voters and 27.6% solid.

Figure 5. Who the Census surname list has no opinion about, by group and by size. Each column is one group, as wide as that group's share of all 49,771 registered voters. The solid block at the top of each column is the share of that group with no surname match, and therefore no score at all. The white column is the widest, at 50.9% of voters, and only 6.6% solid. The Hispanic column is 6.4% of voters and 27.6% solid. **The rule that decides who gets scored is not neutral with respect to who they are.**

The 3,787 voters whose surname is not in the Census list are **23.3% Hispanic, against 6.4% among all registered voters** — over-represented by a factor of about 3.6. Asian voters are over-represented too.

So the Hispanic accuracy reported above was measured on the Hispanic voters with *common* Hispanic surnames. The hard cases were removed before the scoring began, by a rule that looks purely technical. **Every accuracy figure in this document is conditional on the method having had an opinion in the first place.**

13 Everything, against the floor

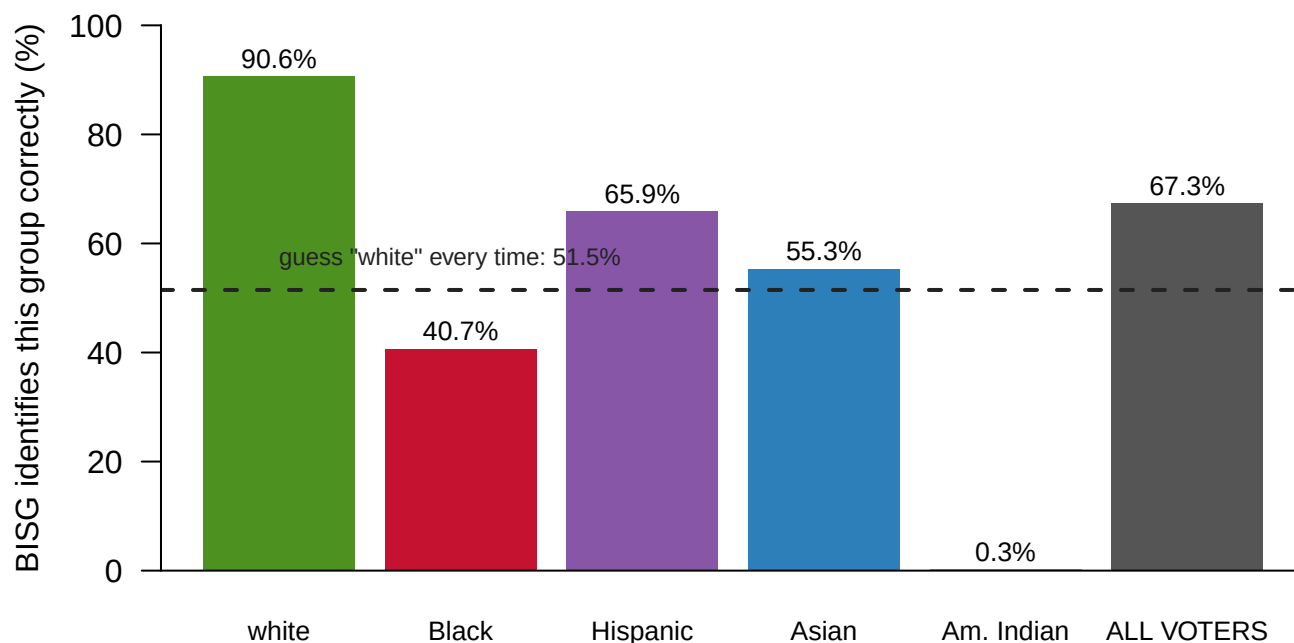


Figure 6. Every accuracy in this chapter, against the number available for free. How often BSG identifies each group correctly, with the whole scored electorate on the right. The dashed line is 51.5% – what guessing “white” every time collects. 2 of the five groups sit below it, and so would the overall bar if the county were not 51.5% white to begin with.

14 Should courts stop using it?

That is not what this evidence supports, and the temptation should be resisted. The defensible statement is narrower: an error rate this uneven should be reported alongside any estimate derived from it, and usually is not. A report saying “BSG was used, with 67% accuracy” is accurate and useless.

Why is it so bad for Black voters? Because the surnames are shared. The common American surnames are held by Black and white families alike: Williams, Johnson, Smith. So the name carries almost no signal, and that is what the 16.9% surname-only figure is measuring, on 16,841 people. It is not a design flaw and not ill will. The signal is genuinely absent, for historical reasons that are themselves part of what the Voting Rights Act exists to address.

And the improvement is real. Imai, Olivella and Rosenman’s fully Bayesian version, which handles the Census file’s own missingness and adds supplementary name lists, improves accuracy most for exactly the groups the simple version handles worst. Better in degree. The kind of thing it is does not change: a probability about a population, attached to a person, used to decide a case about that person’s district.

The validation itself is unrepresentative. Checking BSG requires a state that records race on registration – a small set, concentrated in the South, and overlapping heavily with the states most often in Section 2 litigation. That is either reassuring, because it is where the method is used, or circular, because it is where the method is used. Both

readings are defensible.

15 What a graded guess can testify to

The evidence here is unusually strong in one respect and unusually local in another.

It is strong because these are records for one real person each, carrying the race that person reported and matched to a census block. That is the rare case where a guess nobody can normally check can be graded at all. And the grading is done group by group rather than in aggregate, which is the entire point: an aggregate number on this problem is a headcount wearing a performance figure's clothes.

It is local because it is one county. Houston County, Georgia, about 51% white and moderately segregated, and nothing here says whether it is typical. The surname list is from 2010 while the registration file is not. 7.6% of voters could not be scored at all, and not at random.

"Any-part Black" is the definition used throughout, following Voting Rights Act practice. The stricter definition would give a different denominator, a different bottom number to divide by, and would move every figure above. These are registered voters rather than residents, so the people who never registered are absent, and they are not a random subset either.

And there are five categories, which puts anybody who is both Hispanic and Black in one box.

Nor is the answer key itself beyond question. **The race on a registration form is also a measurement** — one box, chosen by the voter, flattening the same people every categorical race question flattens. Calling it ground truth is a convenience, not a claim about the world.

So the sentence this chapter will bear is that on 45,984 people whose race is on record, BISG scores 67.3% overall — and that the sentence should not be said without the rest of it. 67.3% overall is 90.6% for white voters and 40.7% for Black voters, against a 51.5% floor available to anyone willing to guess "white" every time. Adding block geography bought 23.8 points for Black voters and cost Asian, Hispanic and white voters. And 7.6% of voters were dropped before scoring, disproportionately Hispanic. The shape of all that generalizes; the specific figures should not be quoted anywhere else.

Three questions stay open, and they are the ones practitioners actually argue about. What threshold of accuracy is enough for a method used in federal court, and whether the answer should depend on what is being proven. Whether probabilities should be carried forward rather than collapsed into labels. Figure 4 suggests a real gain is available there, and nobody agrees on how to carry the uncertainty into the second stage. And what to do about a group the method simply cannot see: American Indian voters are identified correctly 0.3% of the time, and no recalibration reaches them.

The method is not illegitimate and the alternative is not better. What is hard to defend is the practice of handing a court one number. In this county that number is 67.3%, and it is the least informative true statement available about what the method did.

16 Sources

Data

- Houston County, Georgia voter registration with self-reported race and geocoded block assignment, from working data for the 2026 Houston County redistricting matter. **Rows whose race had been filled in by BISG rather than reported by the voter were dropped first**; scoring BISG against BISG would have produced an excellent and meaningless result, so those records are dropped before anything is scored.
- Block population by race: 2020 Census P.L. 94-171 and Demographic and Housing Characteristics tables. https://www2.census.gov/programs-surveys/decennial/2020/data/01-Redistricting_File--PL_94-171/Georgia/ga2020.pl.zip
- Surname probabilities: U.S. Census Bureau, *Frequently Occurring Surnames from the 2010 Census* – for each surname held by at least a hundred Americans, the share of holders in each race category. <https://www2.census.gov/topics/genealogy/2010surnames/names.zip>

The data itself

Every figure above is drawn from these tables. They are plain CSV, they open in a spreadsheet, and they are yours to take further.

`houston_blocks.csv`, `houston_voters.csv`, `census_surnames.csv`

Your turn

None of these is answered above, and every one can be answered from the tables you just opened.

- `houston_voters.csv` gives every voter a surname, a self-reported race and a block. Group by race and count. Which groups are large enough here to say anything about?
- Take a common surname from `census_surnames.csv` and find its holders in `houston_voters.csv`. Compare the national shares against what these voters actually reported.
- `houston_blocks.csv` gives the racial composition of each block. Find the blocks with nobody in them. What does the method do when it lands on an empty block?
- Pick a surname whose national shares are close to even. Work out how confident the method could possibly be about anyone carrying it.

Literature

- Elliott, M. N. et al. (2009). "Using the Census Bureau's Surname List to Improve Estimates of Race/Ethnicity and Associated Disparities." *Health Services and Outcomes Research Methodology* 9(2): 69-83.
- Imai, K. and Khanna, K. (2016). "Improving Ecological Inference by Predicting Individual Ethnicity from Voter Registration Records." *Political Analysis* 24(2): 263-272.
- Imai, K., Olivella, S. and Rosenman, E. T. R. (2022). "Addressing Census Data Problems in Race Imputation via Fully Bayesian Improved Surname Geocoding and Name Supplements." *Science Advances* 8(49): eadc9824.
- *Thornburg v. Gingles*, 478 U.S. 30 (1986).

How the figures were made

Every number and every figure above is computed at render time from the three files kept here and the surname list, by re-running BISG from scratch. That means the sur-

name distribution, the block prior, the posterior, the two fallback rules and the scoring. Nothing is copied from a published accuracy figure, and the confusion matrix, the calibration histograms and the accuracy bars are all different views of the same one run.